Sanaul Haque

Vitor_Meneses_Campos_Reis_Proposal

Course 201

Document Details

Submission ID
trn:oid:::3117:637066819

Submission Date
Oct 10, 2026, 11:24 PM GMT+6

Download Date
Oct 10, 2026, 11:27 PM GMT+6

File Name
Vitor_Meneses_Campos_Reis_Proposal.pdf

File Size
177.9 KB

7 Pages

2,529 Words

15,208 Characters

11% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

17 Not Cited or Quoted 10%
Matches with neither in-text citation nor quotation marks

5 Missing Quotations 2%
Matches that are still very similar to source material

1 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

9% Internet sources
7% Publications
9% Submitted works (Student Papers)

Match Groups

- **17** Not Cited or Quoted 10%
  Matches with neither in-text citation nor quotation marks
- **5** Missing Quotations 2%
  Matches that are still very similar to source material
- **1** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

9% Internet sources
7% Publications
9% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Student papers	American University of Bahrain on 2026-01-04 — 2%
2	Student papers	University of Leeds on 2025-08-25 — <1%
3	Internet	iris.paho.org — <1%
4	Student papers	Universidade de Sao Paulo on 2020-04-14 — <1%
5	Internet	www.mdpi.com — <1%
6	Student papers	Berlin School of Business and Innovation on 2026-06-15 — <1%
7	Publication	Kelmara Mendes Vieira, Taiane Keila Matheis, Ana Maria Heinrichs Maciel. "Risky ... — <1%
8	Student papers	University of Sussex on 2025-08-21 — <1%
9	Internet	idec.org.br — <1%
10	Student papers	National University of Singapore on 2026-04-14 — <1%

11 Student papers
University of Leicester on 2022-04-22 <1%

12 Internet
qa1.scielo.br <1%

13 Internet
seer.imed.edu.br <1%

14 Internet
www.bls.gov <1%

15 Student papers
Independent College Dublin on 2023-04-14 <1%

16 Internet
scholar.archive.org <1%

17 Student papers
FACULTY OF SOCIAL SCIENCES (FSS) on 2026-08-17 <1%

18 Internet
seer.atitus.edu.br <1%

19 Internet
www.bcb.gov.br <1%

20 Student papers
Centro de Investigacion y Docencia Economicas AC on 2026-08-19 <1%

21 Publication
Liping Gao, Yufan Wang. "Understanding price–volume relationships and volume ... <1%

IARCO RESEARCH PROPOSAL

Full Legal Name: Vitor Meneses Campos Reis

Institution: Marian Catholic High School

Category: Junior

Class/Grade/Year: Grade 12

Country: Brazil

Submission Date: October 20, 2026

Registered Email Address: vitormenesescamposreis09@gmail.com

Research Topic: Economy, Instant Payment System, Consumer Habits, Public Policies on Income Distribution, Indebtedness.

New Consumption Habits: The Impact of Pix (Brazil's instant payment method) on Default Rates Among Families Dependent on Public Income-Distribution Policies.

Research Problem

Pix, officially introduced by Banco Central do Brasil (BCB) in November 2020, is a payment system that allows money to be transferred instantly between Brazilian financial institutions [1]. One of its main advantages is how simple it is to use: anyone with an account at a financial institution in Brazil can make a payment from a mobile phone anywhere there is internet access [1]. Individuals are not charged a fee for these transactions, banks are not permitted to charge them for this type of operation, and there is no minimum transaction amount [1]. This technology therefore made an important contribution to financial inclusion [2]. At the same time, this convenience changed the way people consume, particularly among low-income groups and households that depend on government income-distribution programs [3]. The change has also been associated with higher household indebtedness and, consequently, default.

Existing Literature

Recent studies have examined the structural changes brought about by the rapid digitization of Brazil's payment system [1], [2], [5]. Pix was introduced by Banco Central do Brasil during the global COVID-19 health emergency [4] and quickly became the country's main infrastructure for instant transfers of funds [1]. The first studies largely approached Pix as a financial-inclusion tool. They showed that free transactions for individuals, together with interoperability between banks and fintechs, helped reduce long-standing barriers to access to banking services among disadvantaged groups [2]. The scale of adoption is clear in a Banco Central do Brasil institutional report: by the end of 2022, just 24 months after implementation, 133 million people, or 77% of the adult population, had completed at least one transaction using Pix [5].

Behavioral Economics also offers a substantial body of research on the way payment methods can shape behavior and affect financial decision-making [3]. In the work of Costa and Casonato, the term “pain of paying” refers to what is described in Portuguese as “dor do gasto”: the idea that payment methods that are less closely associated with physical money can make spending feel less noticeable to the consumer [3]. When these payment methods are built into mobile devices, this effect can become even more pronounced. Santos and Santos, for example, examine how the ability to transfer money instantly, without time or geographic barriers and with little friction or cost, has changed household budget management by making smaller, more immediate and sometimes impulsive purchases easier [6]. Barreto, Avelino, and França make a similar point, arguing that convenient payment methods and increasingly immediate consumption practices can put household budgets under greater pressure by bringing consumers closer to credit and payment facilitators without the traditional barriers that encourage financial deliberation [7]. Online betting platforms, commonly referred to as “bets” in Brazil, are another factor discussed in relation to indebtedness. By making gambling accessible through smartphones, these platforms can encourage people who are vulnerable to gambling addiction or who have difficulty controlling their spending to use part of their income on them [6].

The broader indebtedness figures also show the scale of the problem. In March 2024, for example, 80.4% of Brazilian households were reported as indebted [6]. Even so, the literature has mainly concentrated on structural determinants and on traditional credit instruments, such as bank

loans [10]. Paula points to high interest rates in Brazil, persistent inflationary pressure on essential goods, and the misuse of credit cards and payroll-deducted loans as historical contributors to household indebtedness [8]. Although newer forms of payment, including Pix Parcelado, the installment-based version of Pix, are mentioned in the literature, the analysis is still largely centered on the formal credit system [8].

Default is another important part of this picture. Here, defaulters are understood as people who have overdue debts or have failed to meet their financial obligations [9], [11]. In August 2026, Brazil had 83.9 million defaulters, representing 51.6% of the adult population, with accumulated debts totaling 592 billion reais [9]. Among these defaulters, 27.3% were linked to bank and credit-card debts, 20.1% to financial debts, 20.7% to utilities such as water, electricity, and gas, and 11.5% to services [9]. Serasa's published data, however, did not directly measure or examine the role of Pix and other instant payment methods in these default figures [9].

Despite the growing literature on digital payments and household debt, there is still a lack of quantitative and econometric research that isolates the direct effect of Pix on household default rates, especially among low-income families [8], [10]. The studies available to date are mostly qualitative or sociological, or they focus on traditional empirical determinants and conventional credit products [8], [10]. This leaves an important gap in the literature: the relationship between Pix, debt creation, and eventual default has not yet been measured directly, particularly in connection with the reduction of the “pain of paying” that may occur when Pix is used [3], [6], [8].

Research Question:

This study investigates the central question: To what extent did the rapid diffusion and transaction velocity of Pix influence default rates among low-income Brazilian households between 2021 and 2025? The question focuses on the behavioral and economic *trade-off* between the digitalization of economic systems, which can bring more people into the formal economy [2], and the possibility that this same convenience may increase financial vulnerability and income disparities among low-income households in Brazil [6], [7].

Methodology:

The study will use a strictly quantitative and econometric approach based on time-series analysis, using aggregated monthly data from January 2021 through December 2025 (a period described as 68 monthly observations) [10], [12].

Econometric Calculation and Data:

To measure the effect of Pix on default in quantitative terms and separate that effect from broader macroeconomic fluctuations, the study will apply a Multiple Linear Time-Series Regression model using a Log-Log (Log-Linear) functional form [12].

The econometric equation is specified as:

$$\ln(\text{default_rate}_t) = \beta_0 + \beta_1 \ln(\text{Pix}_t) + \beta_2 \text{Selic}_t + \beta_3 \text{IPCA}_t + \beta_4 \ln(\text{BolsaFamilia}_t) + \beta_5 \text{Unemployment}_t + \varepsilon_t$$

- $\ln(\text{default rate}_t)$ (Y - Dependent Variable): This is the logarithm of the monthly default

rate at time t [9], [11], [12].

- β_0: This is the constant, or intercept, of the equation [12].
- $\beta_1 \ln(Pixt)$ (X - Independent Variable): This is the logarithm of the monthly volume of Pix transactions [1], [5]. The coefficient β_1 measures elasticity, meaning the percentage change in default associated with a 1% increase in Pix usage [12].
- β_2Selict to β_5Unemployment (Control Variables): These variables account for interest rates, inflation, Bolsa Família income, and unemployment in the same period. Their purpose is to separate these effects so that the β_1 coefficient captures the effect attributed to Pix [8], [10], [12].
- εt: This is the stochastic error term, representing external factors that are not observed in month t [12].

The logarithmic transformation "ln" is used because the variables are measured on very different scales, such as percentages and billions of reais, and because the log form allows changes to be interpreted in percentage terms [12]. To check the statistical validity of the model, the series will undergo Unit Root (ADF) and Cointegration tests, and the standard errors will be adjusted using the Newey-West method [13], [14].

Hypotheses:

1. H_1: The convenience of digital transactions through Pix reduces the "pain of paying" and makes unplanned, fragmented purchases more likely, which would produce a statistically significant coefficient $\beta_1 > 0$ ($p < 0.05$) [3], [6], [12].
2. H_2: The positive effect of Pix volume will be stronger for defaults involving essential utility bills (water, electricity, and gas) than for formal bank-credit debt. The expected pattern would reflect a loss of immediate liquidity that leaves less money available for fixed bills at the end of the month [6], [7], [9], [11].
3. H_3: The positive relationship between Pix and the default rate will remain statistically significant even after the model accounts for the macroeconomic environment (β_2Selic, β_3IPCA, β_4BolsaFamília, β_5Unemployment) [8], [10], [12].

Variables:

1. Dependent Variable (Y - Indebtedness and Default): The monthly default rate (%) and indebtedness indicators for low-income households, defined here as households with income of up to 2 minimum wages [9], [11]. The data will be combined from Serasa Experian's Default and Debt Renegotiation Map and the Consumer Indebtedness and Default Survey (PEIC/CNC) [9], [11]. Together, these sources will provide measures of the share of indebted households, the percentage of households that are actually in default because of overdue bills, the average value of their debts, and the total accumulated debt within each income bracket [7], [9], [11].
2. Independent Variable (X - Adoption and Intensity of Pix Use): The monthly value of financial transactions in Brazilian reais (R$) made by individuals, the average value per transaction, the total number of active users (adopters), and the estimated transaction volume involving beneficiaries of social programs [1], [5]. These data will be obtained publicly through the Time Series Management System (SGS) and the Open Data API of

Banco Central do Brasil (BCB) [1], [5].

3. Control Variables (macroeconomic data): These variables include macroeconomic and social indicators that will be used to separate the effect of Pix from broader changes in the economy [8], [10]:
    - Monthly Selic interest rate (SGS/BCB - Series #11) [8], [10];
    - Monthly inflation measured by the IPCA (Brazil's broad consumer price index) (IBGE/SIDRA) [8];
    - Bolsa Família indicators: total monthly financial disbursement, total number of beneficiary families, and the average benefit received per family (Transparency Portal / MDS) [8];
    - Unemployment rate among the low-income population (Continuous National Household Sample Survey — PNAD Contínua/IBGE) [10].

Limitations:

It is not possible to conduct a 100% up-to-date study because some sources only release their data annually or biennially. For this reason, the study was limited to data available through the end of 2025. Since the research has a nationwide scope, it also relies heavily on information collected by government organizations. Therefore, the accuracy and reliability of the data collected may or may not be affected by biases or government manipulation, although this is considered a remote possibility given that Brazil is a democratic country.

It is also important to consider that, because these are secondary data, they may no longer be available depending on the internal policies of the respective institutions responsible for them. However, due to the large scale and volume of the data, it would be extremely difficult and costly for an independent organization or group to collect such information with the same level of accuracy, both because of the scale involved and because some of the data are obtained from private databases maintained by government and financial institutions.

Ethical Considerations and Analytical Integrity

Although standard human subject ethics protocols do not directly apply due to the exclusive reliance on secondary, publicly available, and aggregated data, research ethics remain central to this study. Given that the analysis addresses politically sensitive socioeconomic issues—specifically public income-redistribution policies in Brazil—the econometric modeling and interpretation of results will be conducted with strict objectivity, methodological transparency, and political impartiality.

Expected Results and Conclusion:

If the econometric model produces a positive and statistically significant coefficient for Pix ($\beta_1 > 0$, with $p < 0.05$), the results would indicate that the expansion of Pix was associated with a higher default rate among low-income households even after accounting for the macroeconomic variables [3], [6], [8], [12]. To examine the relationship more directly across types of debt, the model should also be estimated separately for each category of default, comparing utilities with bank credit [9], [11]. It will also be necessary to test whether the Pix coefficient is larger in the regression for essential utilities ($\beta_1$utilities > $\beta_1$banks). A larger coefficient in that regression

would be consistent with the idea that fragmented spending reduces the liquidity available in the household budget [3], [6], [7], [11].

If the data support the hypothesis, the study will have both statistical and practical relevance. Statistically, it would provide evidence of a correlation between the use of instant payments and increased default among people who depend on income-distribution programs [12]. From a practical perspective, such a relationship could inform public policies aimed at limiting the contribution of payment methods to growing indebtedness and default, with the broader objective of reducing income inequality in Brazil [2], [5], [8].

References:

[1] Banco Central do Brasil, "About Pix," Banco Central do Brasil, Brasília, DF, 2020. [Online]. Available: https://www.bcb.gov.br/estabilidadefinanceira/pix-sobre. [Accessed: Sep. 19, 2026].

[2] I. P. O. S. Rimonato and J. P. dos Santos, "PIX: technological solution of financial inclusion," Research, Society and Development, vol. 10, no. 13, 2021. [Online]. Available: https://d1wqtxts1xzle7.cloudfront.net/73392269/18705-libre.pdf. [Accessed: Sep. 18, 2026].

[3] A. F. A. da Costa and L. Casonato, "The influence of payment methods on willingness to pay: an approach to Pix from Behavioral Economics," in Proceedings of the Regional Economics Meeting of ANPEC Sul, 2026. [Online]. Available: https://www.anpec.org.br/sul/2026/submissao/files_I/i6-685771d2199320c73c7f6935be300157.pdf. [Accessed: Sep. 20, 2026].

[4] Pan American Health Organization (PAHO), "History of the international COVID-19 emergency," Pan American Health Organization, 2020. [Online]. Available: https://www.paho.org/en/node/79443. [Accessed: Sep. 20, 2026].

[5] Banco Central do Brasil, "Pix Management Report – 2023," Banco Central do Brasil, Brasília, DF, Institutional Report, 2023. [Online]. Available: https://www.bcb.gov.br/content/estabilidadefinanceira/pix/relatorio_de_gestao_pix/relatorio_gestao_pix_2023.pdf. [Accessed: Sep. 20, 2026].

[6] F. R. dos Santos and F. C. A. S. dos Santos, "Consumption, credit, and everyday life: instant payments (Pix) and contactless payments and new practices for income management in Brazil," Caminhos de Geografia, vol. 24, no. 93, pp. 18–35, Jun. 2023, doi: 10.14393/RCG249365642. [Online]. Available: https://doi.org/10.14393/RCG249365642. [Accessed: Sep. 20, 2026].

[7] F. A. Barreto, P. Avelino, and J. M. S. de França, "The cellphone and over-indebtedness of Brazilian families," Fundação Getulio Vargas (FGV), 2023. [Online]. Available: https://seer.ufu.br/index.php/caminhosdegeografia/article/download/65642/36155. [Accessed: Sep. 20, 2026].

[8] M. A. de Paula, "Indebtedness of Brazilian families and Pix," Undergraduate Thesis, Pontifícia Universidade Católica de São Paulo (PUC-SP), São Paulo, SP, Brazil, 2023. [Online]. Available: https://repositorio.pucsp.br/bitstream/handle/45738/1/TCC%20Endividamento%20das%20famili

as%20brasileirase%20o_MARIA%20APARECIDA%20DE%20P.pdf. [Accessed: Sep. 20, 2026].

[9] Serasa Experian, "Brazilian Default and Debt Renegotiation Map – August 2026," Serasa Experian, São Paulo, SP, Research Report, Aug. 2026. [Online]. Available: https://www.serasa.com.br/limpa-nome-online/blog/mapa-da-inadimplencia-e-renogociacao-de-dividas-no-brasil. [Accessed: Sep. 20, 2026].

[10] M. Daros and N. G. M. Pinto, "Default in Brazil: an analysis of empirical evidence," RAIMED – Revista de Administração IMED, vol. 7, no. 1, pp. 208–229, 2017, doi: 10.18256/2237-7956/raimed.v7n1p208-229. [Online]. Available: https://doi.org/10.18256/2237-7956/raimed.v7n1p208-229. [Accessed: Sep. 20, 2026].

[11] National Confederation of Trade in Goods, Services and Tourism (CNC), "Consumer Indebtedness and Default Survey (PEIC)," CNC, Brasília, DF, Monthly Statistical Report, 2024–2026. [Online]. Available: https://portaldocomercio.org.br/pesquisas/peic. [Accessed: Sep. 20, 2026].

[12] J. M. Wooldridge, *Introduction to Econometrics: A Modern Approach*, 7th ed. São Paulo, SP, Brazil: Cengage Learning, 2020.

[13] D. A. Dickey and W. A. Fuller, "Distribution of the estimators for autoregressive time series with a unit root," *Journal of the American Statistical Association*, vol. 74, no. 366, pp. 427–431, 1979.

[14] W. K. Newey and K. D. West, "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix," *Econometrica*, vol. 55, no. 3, pp. 703–708, 1987.